Tom Whipple

Experienced AI/Software Engineering Leader · Boulder, Colorado · 612-220-1465 · mail@tomwhipple.com

EXPERIENCE

Software/AI Engineer — Freelance Mar 2025 – Present

Luna Physical Therapy · Remote

- Developed tools to analyze performance of call center agents (human and AI) to ensure conversations met standards for excellent customer service, including consistency with a predefined knowledge base; used human feedback to fine-tune models predicting manual call QA scores.
- Built ETL tools to import/export data between AWS stack and healthcare formats (QRDA1) using Python & Elixir.
- Provided SME advice to business stakeholders.

Software Engineering Manager Apr 2024 – Dec 2024

Numa · Remote

- Led a team responsible for building automation tools for auto dealer service departments, including a voice conversational scheduler.

Software Engineer Sep 2022 – Apr 2023

Meso · Remote

- Built a crypto payment on-ramp using Go. Led the architecture for the risk system and participated in overall system architecture and API design.

Senior Software Engineering Manager Jul 2012 – Jun 2020

PayPal (Braintree) · Austin, TX (Hybrid)

- Led an organization of 20 across three teams, responsible for all alternative payment method processing (PayPal, Venmo) within Braintree.
- Grew transaction volume from the low millions to more than $2B/month; collaborated across the company to reduce API errors and lower spurious declines.
- Grew a team of 4–5 engineers to 20 through external hiring and internal talent acquisition.
- Presented to executive leadership to drive company-wide focus on API quality; architected solutions resulting in two patents.
- Served as Austin site lead for Braintree, bridging the gap between PayPal enterprise and Braintree startup cultures.

Principal Software Engineer Jul 2011 – Jul 2012

card.io · San Francisco, CA · Acquired by PayPal 2012

- Technical lead for the Android SDK and fraud control; contributed to the computer vision backend and iOS SDK using Java, Objective-C, and Python.

Founder Aug 2009 – Jul 2011

Smartovation Technologies · Ventura, CA

- Founded a contract development firm specializing in mobile apps; clients included Sprinkles Cupcakes and Emirates Airlines.

Computer Engineer Sep 2007 – Aug 2009

NAVAIR · Point Mugu, CA

- Developed HiL test capability for an airborne ultra-wideband digital RF receiver; Technical POC for two SBIR projects involving digital wideband RF receivers.
- Developed a maximum likelihood algorithm for classifying RF signals in Matlab; wrote real-time control software and a VxWorks device driver in C++.

Graduate Technical Intern May – Aug 2006

Intel · Portland, OR

- Designed and created an open source application test suite for the 1.0 release of Linux utility xdg-utils (freedesktop.org Portland project).

Software Engineer 2000 – 2003

Nistevo · Twin Cities, MN

- Built a Makefile-driven build/test system; developed a new QA test framework; served as lead developer for business reporting; created automated monitoring and recovery tools in Perl.

System/Network Administrator 1997 – 2000

Znet Telecom · Minnesota

- Installed and maintained web, mail, FTP, NNTP, and dialup services on Linux; migrated legacy Windows servers to Linux; built customer support automation tools.

EDUCATION

MS, Computer Science 2005 – 2007

University of Minnesota

- Primary area of study: machine learning and computer vision; additional coursework in robotics and embedded systems.

BS, Mathematics 2003 – 2005

University of Minnesota

- Member of the 2nd place 2005 Solar Vehicle Project team; served as lead race strategist and embedded software developer.

PATENTS

Systems and methods for authenticating a user based on a computing device Issued Jan 2, 2018 · US9858405

- Authentication of a user based on a "device fingerprint" stochastically predicted from a large number of attributes inherent to the device, browser, or app.

Conditional Transaction Pre-Approval Filed Jan 31, 2022 · US20220156750

- Use machine learning to predict the outcome of a payment authorization request, in order to respond more quickly and/or hide an outage in a downstream system.